



Principles of Block Cipher (GFG Based)



1. Block Size

- Data is divided into **fixed-size blocks**
 - Should not be too small (less secure)
 - Common sizes: **64-bit, 128-bit** ([GeeksforGeeks](#))
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2. Key Size

- Determines **security strength**
 - Larger key → more secure
 - Small key → easy to break
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3. Number of Rounds

- Encryption is done in **multiple rounds**
 - More rounds → stronger security
 - Each round performs similar operations
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4. Substitution (Confusion)

- Replaces data values
 - Hides relationship between **key and ciphertext**
 - Makes encryption complex
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5. Permutation (Diffusion)

- Rearranges bits of data
 - Spreads effect of one bit across many bits
 - Prevents pattern detection ([GeeksforGeeks](#))
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6. Function Design (Round Function)

- Core part of cipher (called **F-function**)
 - Combines substitution + permutation
 - Applied in every round
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7. Key Scheduling

- Generates **different subkeys** from main key
 - Each round uses a different key
 - Improves security
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Short Conclusion (Write in Exam)

Block cipher design principles include block size, key size, number of rounds, substitution, permutation, and key scheduling to ensure strong, secure, and efficient encryption.



Easy Memory Trick

👉 “**B K R S P F K**”

- B → Block size
 - K → Key size
 - R → Rounds
 - S → Substitution
 - P → Permutation
 - F → Function
 - K → Key schedule
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If you want, I can make **ultra-short 2-mark answer** or **10-mark detailed answer with explanation flow** 👍